



Finance Course and Player Guide

Everything in Sour Milk Logistics, explained: how to play, the rules the world runs on, and the finance each part of the game teaches, from break-even to futures.



Part 1 · Play

Getting started, every screen, both tutorials and four ways to make money.

Part 2 · Learn

Eighteen finance modules, each with the game rule, the theory, a formula, a worked example and something to try.

Part 3 · Compete

Online matches: the market preview, host trends, win conditions and a class session plan.

Part 4 · Practise

A course map, 24 practice questions with answers, and a glossary.

Play free in a browser: **logistics.sourmilkstudioos.com** · Based on the game as published on 26 October 2026 · Contact: sourmilkstudios1@gmail.com

START HERE

How to use this guide

Sour Milk Logistics puts you in charge of a small shipping company inside a working world economy: ten countries, 79 listed companies, central banks, commodity prices that follow real stock levels, and a share market that follows real earnings. Nothing in the numbers is faked: a company earns what it actually sells, and a late ship in one port can stop a factory three countries away.

For players

Read Part 1 next to the game. It covers every screen and both guided tutorials, then shows four ways to make money.

For students

Play the tutorial first, then work through the modules in Part 2. Each one ends with something to try in the game. Part 4 has practice questions with answers.

For lecturers and clubs

Part 3 turns an online match into a class session. The course map in Part 4 shows which module fits which course.

What is new in this edition

The first edition (5 October 2026) had ten modules. The game has grown a lot since, so this edition is rewritten and has eighteen. New in the game, and new in this guide:

- **Factories you run by hand** (Normal mode): worker bots with moods, modules that wear out and break, a warehouse, marketing by country, a supply chain and a 12-month financial model. *Modules 5–8, 10.*
- **Live company reports** with a health grade and a “where every $100 goes” bar. *Module 9.*
- **Live trading**: share prices move hour by hour, candlestick charts, indicators, drawing tools and your entry price highlighted. *Modules 12–13.*
- **Futures**: long or short, 1× to 10× leverage, margin and liquidation. *Module 14.*
- **Secured bank loans** with collateral, monthly payments, margin calls and repossession. *Module 15.*
- **Market rules** against dumping and pump-and-dump, with warnings, fines and bans. *Module 16.*
- **The company rating** (AAA to D) that can decide online matches. *Module 18.*
- **Online matches** with a 30-day market preview, a countdown, host-chosen market trends and four win conditions. *Part 3.*

Contents

Part 1 · Play	1.1 Getting started · 1.2 The screens · 1.3 The two tutorials · 1.4 Four ways to make money
Part 2 · Learn	1 Cost structure and break-even · 2 Revenue and the time value of a promise · 3 Trading and arbitrage · 4 Supply, demand and prices · 5 Factory unit economics · 6 Operations: bottlenecks, maintenance and people · 7 Supply chain risk · 8 Marketing and pricing · 9 Reading the company report · 10 Forecasting: the financial model · 11 Capital budgeting and valuation · 12 Shares and the stock market · 13 Technical analysis · 14 Futures and leverage · 15 Debt, collateral and default · 16 Market integrity · 17 Interest rates, inflation and the cycle · 18 Measuring performance: the company rating
Part 3 · Compete	3.1 How an online match runs · 3.2 Host controls · 3.3 A 90-minute class session
Part 4 · Practise	4.1 Course map · 4.2 Practice questions · 4.3 Answer key · 4.4 Glossary



PART 1

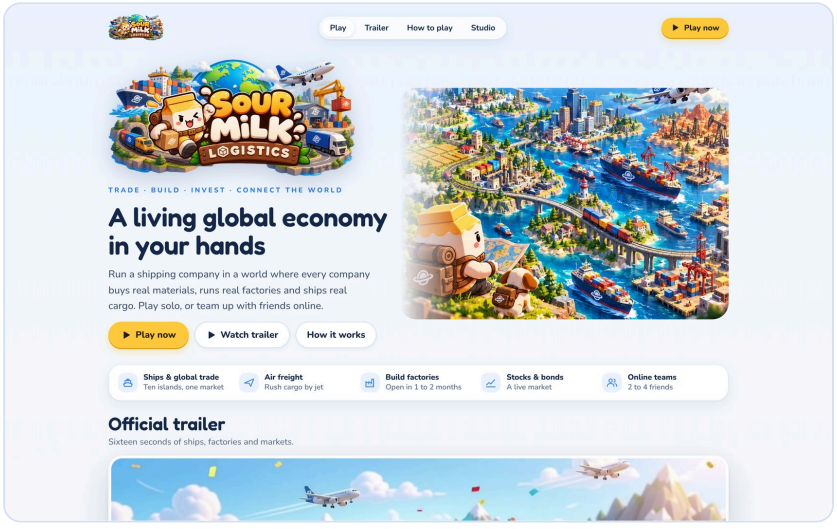
Play

Getting started, every screen, both tutorials and four ways to make money.

1.1 Getting started

The game is free and runs in a web browser at **logistics.sourmilkstudioos.com** on a computer or a phone. Nothing needs to be installed, and it saves itself every 30 game days. Before a new game you make three choices.

CHOICE	OPTIONS	WHAT TO PICK
Mode	Normal or Pro	Normal first. Your own factories are simple to run. Pro adds the full stock and bond market, central-bank detail, company takeovers and a factory you plan in detail.
Starting capital	Founder $12.48M · Investor $1B · Tycoon $60B	Founder is the intended game. Bigger starts let you skip ahead.
Tutorial	Play it or skip it	Play it the first time. It takes about two minutes.
Play online	Create or join a match	See Part 3. Teams of 1 to 4 players share one world.

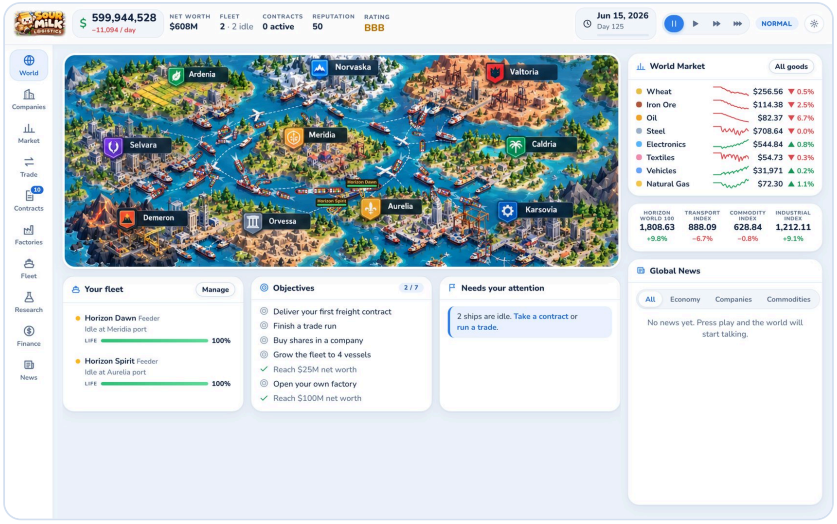


The title screen: Normal and Pro, starting capital, the tutorial switch, the trailer and Play online.

The clock

The game starts paused. Press Play (or the space bar). At Play speed one day lasts 12 seconds; there are two faster speeds. The gold bar under the date shows today passing, and share prices move hour by hour within the day. In an online match the server keeps the clock and nobody can pause.

1.2 The screens



The World screen: the living map, your fleet, the leaderboard or objectives, commodity prices, market indices and the news.

SCREEN	WHAT YOU DO THERE
World	Watch the map. Every ship that moves is a real shipment between companies. Click an island for its port, prices and companies.
Companies	Browse all 79 companies; open one to see its factory, suppliers, finances and shares.
Market	Commodity prices, stocks, **Live trading** (live prices and futures) and, in Pro, bonds and central banks.
Trade	Buy goods in one port with your own ship and sell them in another.
Contracts	Carry other companies' cargo for a fee, by a deadline.
Factories	Found your own company and open the companies you run.
Fleet	Your ships, their life bars and the shipyard.
Research	Six projects that permanently improve your fleet, trading or construction.
Finance	Your company rating, net worth, bank loans, the market rules, your shareholdings and the ledger.
News	World, economy, company and commodity news that moves prices.

The gear (top right) holds saving, the day length, both tutorials, the advisor and **How to play**, a one-page summary of this guide.

1.3 The two tutorials

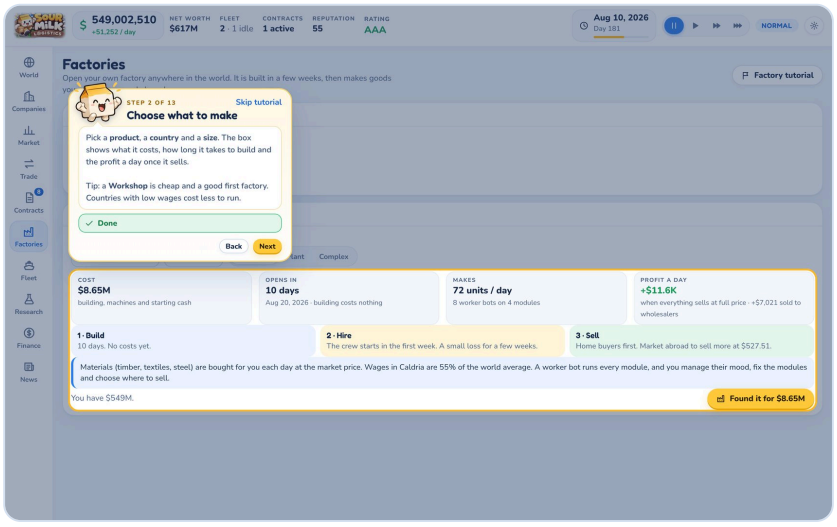
The studio's milk-carton mascot guides both. A spotlight moves across the real game and some steps wait for you to act. You can go back, skip a step or skip the tour.

The guided tour (13 steps, starts with a new game)

STEP	WHAT IT SHOWS	FINANCE IDEA
1–3	Welcome, your money, the clock (press Play)	The firm as a going concern; cash versus income
4–5	The living world and a country (click an island)	Markets, ports and central banks
6–7	Contracts: take your first one	Revenue earned on delivery
8	Trade: the best buy–sell spreads from your port	Arbitrage
9–10	Inside a company: the factory and the Why? chain	Operations and root-cause analysis
11	Invest (Normal) or Own it (Pro)	Equity, dividends and control
12	Build your own factory	Capital budgeting
13	You are ready: objectives and the advisor	Setting goals

The factory tutorial (13 steps, from the Factories screen or the menu)

It takes you from founding a factory to selling abroad: choose a product, country and size; construction (in single player you can skip to opening day); click a worker bot and talk to it; modules, health and pace; the warehouse and selling to wholesalers; where the money goes; the "What to do now" box; marketing in a second country; pay and perks; growing the factory.



The factory tutorial on the Factories screen: choose what to make, then found it.

1.4 Four ways to make money

1 · Contracts

Shippers pay you to carry cargo by a deadline. Deliver on time for full pay and reputation. *Modules 1–2.*

2 · Trade

Buy a good where it is cheap, sail, sell where it is dear. The Trade screen shows the profit after fuel. *Modules 3–4.*

3 · Factories

Found your own producer. Built in 10–20 days, then it earns every day. Half its profit is paid to you each quarter. *Modules 5–11.*

4 · Markets

Buy shares for dividends and gains, or trade futures, long or short. *Modules 12–14.*

Objectives

OBJECTIVE	WHAT IT TEACHES	MODE
Deliver your first freight contract	Earning revenue on time	Both
Finish a trade run	Buying and selling on your own account	Both
Buy shares in a company	Becoming an investor	Both
Grow the fleet to 4 vessels	Reinvesting profit in capacity	Both
Reach $25M net worth	Building equity	Both
Open your own factory	A long-term capital project	Both
Become a major shareholder (5%)	Voting rights	Pro
Take control of a public company (50%)	Corporate control	Pro
Reach $100M net worth	Compounding	Both

Research

PROJECT	COST	DAYS	EFFECT
Efficient Hulls	$2.5M	60	Fleet fuel use −15%
Fast Turnaround	$3.0M	75	Loading and unloading 40% faster
Weather Routing	$4.0M	90	Ships 10% faster, half the storm delay
Key Account Desk	$3.5M	80	Freight contracts pay 8% more
Trade Analytics	$2.0M	50	Price board for every port; trading slippage −30%
Lean Construction	$6.0M	120	New factories cost 12% less and finish 20% faster

Research is a classic investment decision: pay now for a permanent gain later. Module 11 shows how to compare them.



PART 2

Learn

Eighteen finance modules. Each states the rule the game uses, explains the finance behind it, works an example with the game's own numbers and ends with something to try. Modules marked **Pro** need Pro mode for the full version.

MODULE 1

Cost structure and break-even

Supports: Managerial Accounting · Corporate Finance · Operations

IN THE GAME

Ships cost money every day, even in port, and burn fuel when they sail. Four classes are sold at the shipyard.

SHIP	CAPACITY	PRICE	FIXED COST / DAY	PER TEU / DAY	SPEED	FUEL
Feeder	400 TEU	$4.2M	$5,500	$13.75	150 nm/day	0.45 bbl/nm
Panamax	1,500 TEU	$13.5M	$13,000	$8.67	140 nm/day	1.3 bbl/nm
Ultra Large	5,000 TEU	$24M	$27,000	$5.40	125 nm/day	3.4 bbl/nm
Cargo Jet	60 TEU	$9M	$9,000	$150.00	700 nm/day	0.5 bbl/nm

THE FINANCE

A **fixed cost** is paid whatever you do (the ship's daily cost); a **variable cost** rises with activity (fuel per nautical mile). Bigger ships have lower fixed cost *per unit of capacity*: economies of scale. But scale only pays when the capacity is used, so the real question is the **break-even**: how much revenue covers all the costs of a trip.

FORMULA

Trip cost = fixed cost per day × days at sea and in port + distance × fuel per nm × fuel price

Break-even revenue = trip cost · Profit = revenue − trip cost

WORKED EXAMPLE

A 1,000 nm trip with fuel at about $150 a barrel. Days ≈ 1,000 ÷ speed + 2 days in port.

Feeder: 8.7 days × $5,500 = $47,850 fixed + 1,000 × 0.45 × $150 = $67,500 fuel → **$115,350**.
Panamax: 9.1 days × $13,000 = $118,300 + 1,000 × 1.3 × $150 = $195,000 → **$313,300**.

The Panamax costs 2.7× as much but carries 3.75× the cargo. On a full load it costs $209 per TEU against $288 for the Feeder; half empty, it is worse. Your two starting Feeders cost $11,000 a day, or $330,000 a month, before they earn anything.

TRY IT IN THE GAME

Compare a contract's pay with the trip cost before you accept it. Watch the "per day" figure under your cash: idle ships pull it down.

MODULE 2

Revenue and the time value of a promise

Supports: Financial Accounting · Business Law · Service Operations

IN THE GAME

A contract pays when the cargo arrives. If it is late, pay drops 6% for each day late, to a floor of 40%. Every on-time delivery adds 1.5 reputation points; every late one costs 3. The Key Account Desk research raises contract pay by 8%.

THE FINANCE

Revenue is **recognised** when the service is delivered, not when you accept the job. Deadlines turn time into money: a late delivery is a breach that carries a penalty, and reputation is an intangible asset that is slow to build and quick to lose. In the game, reputation also feeds your company rating (Module 18).

FORMULA

Pay received = contract pay × max(0.40, 1 − 0.06 × days late) × (1.08 with Key Account Desk)

WORKED EXAMPLE

A $200,000 contract delivered 3 days late pays $200,000 × (1 − 0.18) = **$164,000**, and reputation falls 3 points. Delivered 10 days late it pays the floor, 40% = $80,000. A Feeder that arrives on time on a $200,000 job worth $115,350 in trip costs earns $84,650; three days late it earns $48,650, 43% less.

TRY IT IN THE GAME

Before you assign a ship, compare its arrival day with the deadline on the contract card. Urgent air cargo needs a Cargo Jet.

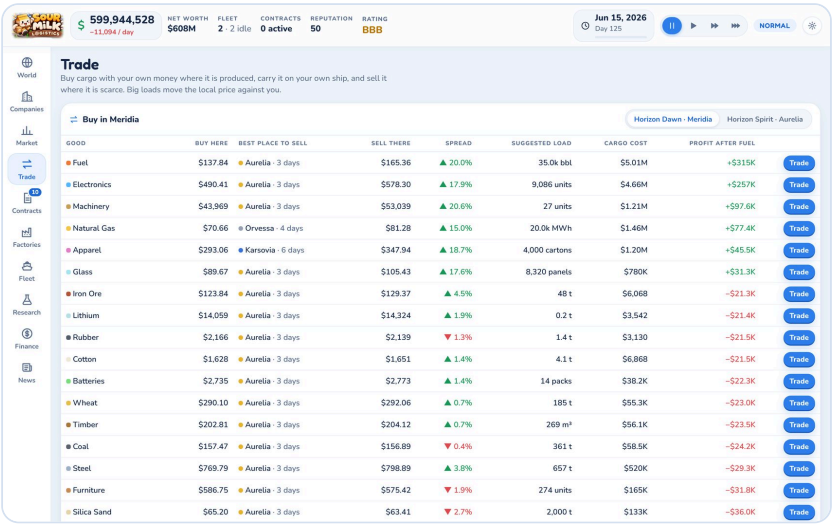
MODULE 3

Trading and arbitrage

Supports: Investments · International Trade · Market Microstructure

IN THE GAME

On the Trade screen your idle ship can buy a good in its port and carry it to another. Exporting ports sell cheap; importing ports pay more. The table shows the best destination, the spread and the profit after fuel.



The Trade screen: what your ship can buy here, where to sell it and the profit after fuel.

THE FINANCE

Arbitrage is buying in one market and selling in another where the price is higher. It is never free: you pay transport and time, and a big order moves the price against you. That is **slippage** (market impact): the deeper the market, the less a trade moves it. Online, your trades also move the local price for every other team, and the market rules (Module 16) stop anyone from flooding a port.

FORMULA

Slip = 0.35 × quantity ÷ port depth (×0.7 with Trade Analytics)

Buy price = local price × (1.01 + slip) · Sell price = local price × max(0.30, 0.99 − slip)

Profit = (sell − buy) × quantity − fuel − days × ship fixed cost

WORKED EXAMPLE

A good costs $100 in port A (daily trade 10,000 units) and $120 in port B (8,000 units). Buying 2,000 units: slip = 0.35 × 2,000 ÷ 10,000 = 0.07, so you pay $108. Selling them in B: slip = 0.0875, so you get $120 × 0.9025 = $108.30. The $20 gap shrinks to $0.30. Buying 500 units instead: pay $102.75, receive $116.18, a spread of $13.43 a unit. **Smaller trades keep more of the spread.**

TRY IT IN THE GAME

Open a trade and move the quantity slider: watch the price per unit get worse as the load grows. The suggested load already balances size against slippage.

MODULE 4

Supply, demand and prices

Supports: Microeconomics · Operations · Commodity Markets

IN THE GAME

Every good has a world price. When producers cannot keep up, **stock cover** (days of stock left) falls and the price climbs; when they are overstocked it falls. Each port's local price differs from the world price depending on whether that country exports or imports the good. News (a float-bath failure, a port strike, an oil shock) moves supply, and prices follow.

THE FINANCE

Prices are a signal of scarcity. Inventory cover links supply and demand: low cover means buyers compete for little stock, so the price rises until demand falls or supply recovers. A factory's costs follow its material prices, which is why a rise in steel or timber shows up in your own unit cost (Modules 5 and 7).

FORMULA

Stock cover (days) = stock on hand ÷ daily demand

Price pressure rises as cover falls below its normal level

WORKED EXAMPLE

Steel has normal cover of 20 days. A blast-furnace failure cuts output and cover drops to 12 days: steel is scarce, and its price rises over the following weeks. A furniture factory using 0.02 t of steel per unit barely notices; a vehicle maker using 1.6 t per car sees its unit cost jump.

TRY IT IN THE GAME

Open Market → Commodities and click a good: you see its cover, who makes it, who uses it and its local price in every port.

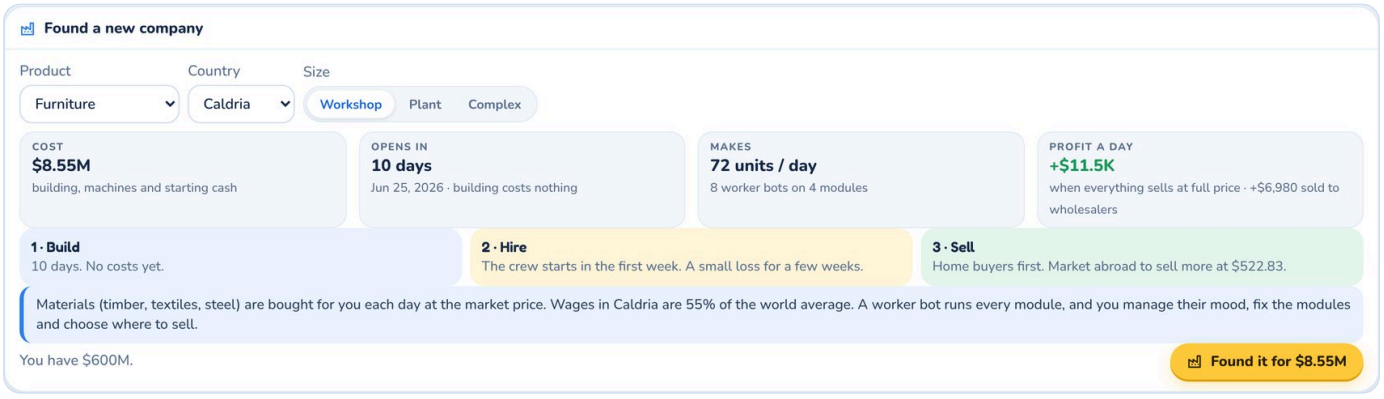
MODULE 5

Factory unit economics

Supports: Managerial Accounting · Cost Accounting · Entrepreneurship

IN THE GAME

In Normal mode you choose a product, a country and a size (Workshop, Plant or Complex). The founding screen shows the cost, the days to open (10, 15 or 20), the output and the expected profit a day: home buyers at full price, the rest to wholesalers, as a yearly average that swings with the season. Construction costs nothing to run; the crew is hired in the first week.



Founding a furniture Workshop in Caldria: cost, opening day, output, profit a day, and the three stages Build, Hire, Sell.

Each unit sells at the market price in your markets, or at **88%** of it to wholesalers. Materials are bought each day at world prices (never more than 72% of the selling price). Energy is about 4% of the value added. Wages, upkeep and marketing are fixed costs per day.

THE FINANCE

The **contribution margin** is what each unit adds after its variable costs; it pays the fixed costs first, and only then makes profit. The **break-even volume** is the number of units that covers the fixed costs exactly. Below it you lose money; above it, every unit adds its full margin to profit. This is **operating leverage**: high fixed costs make profit swing more as volume changes.

FORMULA

Contribution margin per unit = price − variable cost per unit

Break-even units per day = fixed costs per day ÷ contribution margin

Profit per day = units × contribution margin − fixed costs

WORKED EXAMPLE

From the furniture Workshop above, after a month of trading (its financial model, Module 10): price $527.64, variable cost $285.85 → margin **$241.78**. Fixed costs $7,378 a day → break-even = 7,378 ÷ 241.78 = **30.5 units a day** at full price. The line makes 72 a day. Selling 50 at full price and 22 to wholesalers: 50 × $241.78 + 22 × ($527.64 × 0.88 − $285.85) − $7,378 = $12,089 + $3,926 − $7,378 = **$8,637 a day**.

TRY IT IN THE GAME

Open Financials → Financial model on your factory and read its break-even. Then raise marketing and watch more units sell at full price.

MODULE 6

Operations: bottlenecks, maintenance and people

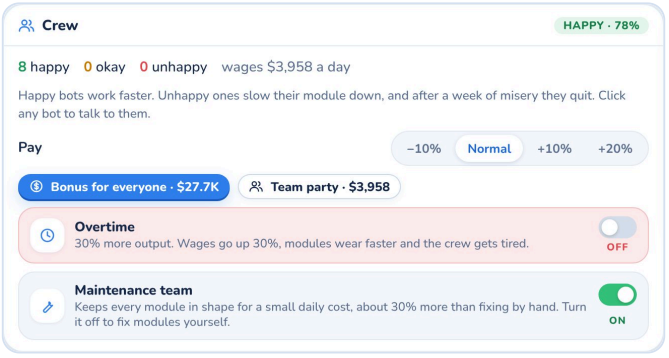
Supports: Operations Management · Human Resource Management · Organisational Behaviour

IN THE GAME

Each factory has 4 to 7 **modules** (for furniture: Cutting, Joinery, Upholstery, Finishing). Each has worker bots (2, 3 or 4 per module by size, one more per upgrade level) and a health bar. A module's pace = health × crew present × mood. **The slowest module sets the pace of the whole line.**



Modules with health, pace and their worker bots. The faces show mood.



Pay level, bonus, team party, overtime and the maintenance team. Click any bot on the factory picture to talk to it.

THE FINANCE

The **theory of constraints**: a line runs only as fast as its bottleneck, so improving any other step adds nothing. Maintenance is a trade-off between planned cost (servicing) and unplanned cost (breakdowns that stop everything). People are a capital, not just a cost: morale changes productivity and staff turnover is expensive.

- Modules lose about 0.55% health a day (faster on overtime or with an unhappy crew) and can break below 60%. The maintenance team keeps them in shape for about 30% more than fixing by hand.
- Pay −10% to +20%; a bonus costs 7 days of wages, a party one day (monthly). Overtime: +30% output, +50% wages, modules wear almost twice as fast. Higher pay keeps the crew happy and slightly faster, but it is a trade-off, not a free win. A bot below 30% mood for six days quits.

FORMULA

Line output = full speed × min(pace of every module) × hiring ramp × (1.3 on overtime)

WORKED EXAMPLE

Four modules at pace 101%, 98%, 99% and 62% (a worn Joinery). The line runs at 62%: 45 units a day instead of 72. At $241 margin that is about $6,500 of lost contribution a day. A fix costs a fraction of that, so fix the bottleneck first.

TRY IT IN THE GAME

Turn the maintenance team off for a month and fix modules by hand. Compare the maintenance line in your report with the month before.

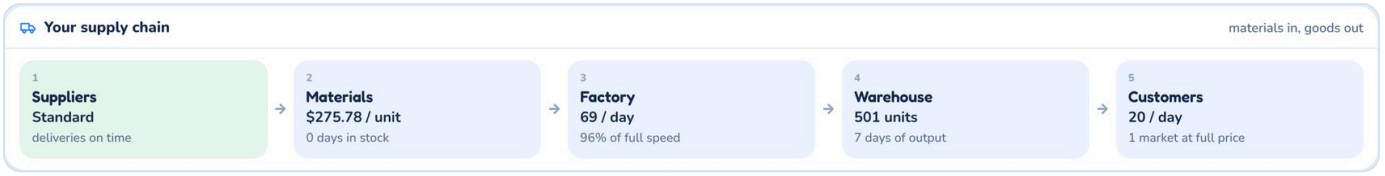
MODULE 7

Supply chain risk

Supports: Supply Chain Management · Risk Management · Operations

IN THE GAME

The **Supply chain** tab shows the flow from suppliers to customers, the materials in one unit and their prices, and two choices.



Suppliers → materials → factory → warehouse → customers.



Budget, Standard or Premium suppliers.

SUPPLIER	MATERIALS COST	SELLING PRICE	LATE DELIVERIES
Budget	−5%	−2%	about 1.5 a month
Standard	normal	normal	about 0.5 a month
Premium	+5%	+2.5%	almost never

A late delivery lasts 2–4 days and slows the line to 40%, unless your **safety stock** (0, 7 or 14 days of materials) covers it. Cheaper materials give a rougher product, so buyers pay a little less; premium materials earn a little more. Each day of safety stock costs 0.12% of a day's materials, every day.

THE FINANCE

Cheaper suppliers raise **expected disruption cost**. The right choice minimises total cost: purchase price + expected loss from disruptions + the cost of holding buffer stock. Safety stock is insurance: you pay a small premium every day to avoid a large, uncertain loss.

FORMULA

Expected disruption cost per month = late deliveries per month × days per delay × output lost per day × margin per unit

Safety stock cost per day = days held × daily material use × 0.0012

Supplier effect per day = materials saved − selling price lost (or the reverse for Premium)

WORKED EXAMPLE

A Workshop using $272 of materials per unit, selling at $527.64 and making 80 a day spends $21,760 a day on materials. Budget saves 5% = $1,088 a day but buyers pay 2% less: 80 × $527.64 × 0.02 = $844, so it nets **$244 a day**. Its 1.5 delays a month × 3 days × 48 units lost (60% of 80) × $241 margin = **$52,000 a month** (about $1,735 a day) of lost contribution, so Budget with no stock is a bad deal. Seven days of safety stock cost 7 × $21,760 × 0.0012 = $183 a day and cover the delays: Budget then nets about $61 a day. Premium costs $1,088 more and earns $1,055 more, almost a wash, with no delays and no stock needed. Standard with no stock quietly loses about $520 a day to its own delays. The lesson: whichever supplier you pick, cover the delays.

TRY IT IN THE GAME

Switch to Budget with no safety stock and watch for the late-delivery alert, then add 7 days of stock and compare.

MODULE 8

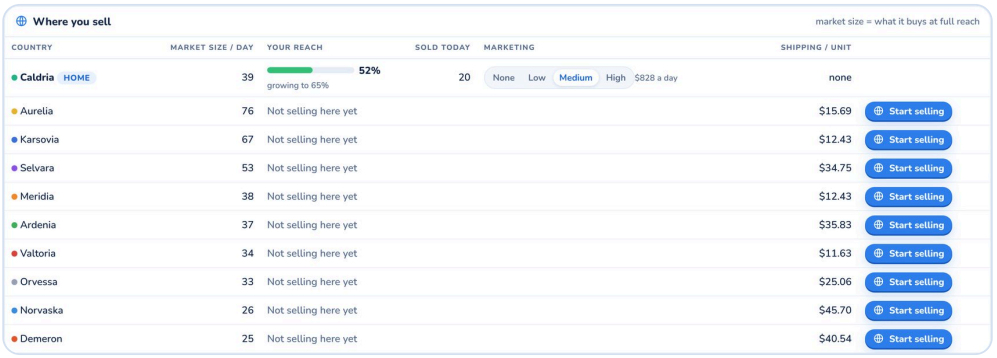
Marketing and pricing

Supports: Marketing Management · International Business · Pricing

IN THE GAME

Every country has buyers for your product. **Reach** is how many of them know you. Marketing grows it each day; with none it slowly fades.

LEVEL	REACH TARGET	GROWTH PER DAY	COST (SHARE OF THAT MARKET'S FULL-REACH SALES)
None	fades	−0.6% a day	free
Low	35%	1.2% of the gap	0.8%
Medium	65%	2.2% of the gap	1.8%
High	100%	3.5% of the gap	3.5%



Sales & marketing: market size, your reach, units sold, marketing level, shipping and gain vs wholesale for every country.

You always sell at home. Exports pay shipping per unit. What markets don't buy goes to the warehouse, and anything older than 7, 14 or 30 days of output is sold to wholesalers, normally at 88% of the price.

Market climate: each product has a busy season, the market drifts over weeks, and a craze or slump hits once or twice a year. Strong markets sell more at full price and wholesalers pay more; weak ones the reverse. The Market climate panel shows the season ahead.

THE FINANCE

Marketing is an investment in a customer base that pays back over time. The **channel trade-off**: full price through your own markets versus a lower, certain price through wholesalers. The **marginal rule**: raise marketing in a country while the extra full-price sales are worth more than the extra cost.

FORMULA

Extra profit from marketing = extra full-price units × (price − wholesale price) − marketing cost − shipping

Wholesale price = 0.88 × market price

WORKED EXAMPLE

A country buys 88 units a day at full reach, and you sell there at Low (35% reach, 31 units). Moving to Medium raises reach to 65% over a few weeks: 26 more units a day sold at full price instead of to wholesalers, each worth $526.79 × 0.12 = $63 more, so $1,640 a day. Medium costs 1.8% of $46,400 = $835 a day against Low's $371: $464 more. Net gain about $1,176 a day, less shipping. The Sales & marketing tab shows this for every country as "gain vs wholesale" per unit: green means marketing there pays.

TRY IT IN THE GAME

Start selling in a second country at Low and watch its reach and "sold today" grow on the Sales & marketing tab.

MODULE 9

Reading the company report

Supports: Financial Accounting · Financial Statement Analysis

IN THE GAME

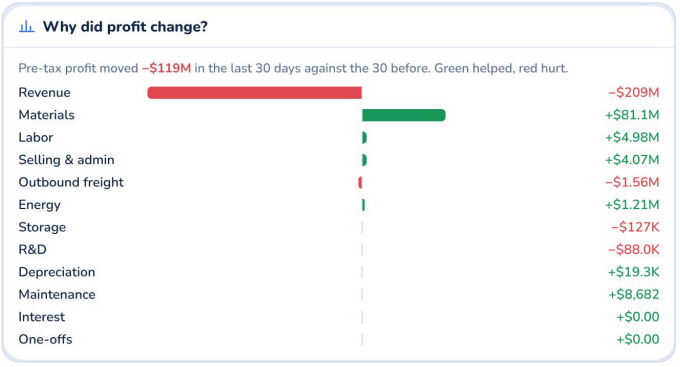
Every company's **Financials** tab starts with a **Live report** that updates each day: a health grade, sales, costs and profit for the last 30 days, three plain-language sentences, and "where every $100 of sales goes". Below it are the quarterly income statement, the balance sheet and "Why did profit change?".



The live report: grade, 30-day sales, costs, profit, today's result and the $100 split.



Your factory's money today, line by line.



Why did profit change? Green helped, red hurt (Pro company).

THE FINANCE

The **income statement** runs from revenue down to net income: materials (cost of goods sold), labour, energy, maintenance, freight, selling and admin, R&D, storage, depreciation, then interest and tax (22% on profit). The **balance sheet** lists what the company owns (cash, inventory, plant) against what it owes (bank debt, bonds); the difference is shareholders' equity. A **common-size** statement (the $100 bar) shows each cost as a share of sales so companies of any size compare. A **profit bridge** explains the change between two periods line by line.

FORMULA

Gross margin = (revenue − materials) ÷ revenue · Operating margin = operating profit ÷ revenue
Net income = (operating profit − interest) × (1 − 0.22) when positive · Equity = assets − debt

WORKED EXAMPLE

The live report above: sales $1.00M, costs $815K, profit after tax $145K, so the net margin is 14.5%. Materials take $57 of every $100, labour $12, maintenance $5, selling $5, energy $2, leaving $19 of profit before tax. The grade is A+ because the margin is above 12%, sales are growing and cash covers 69 days of costs.

TRY IT IN THE GAME

Compare the $100 bar of your factory with a large listed producer of the same good. Where does the big company spend less?

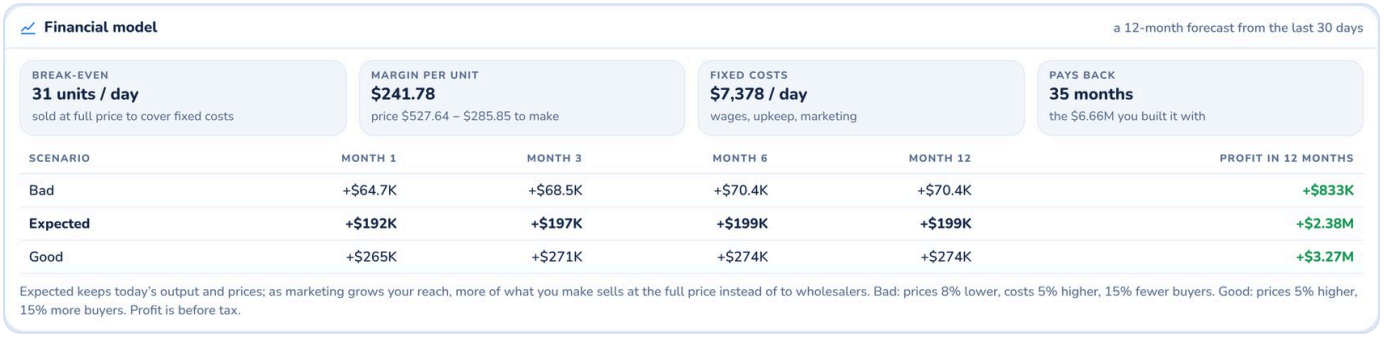
MODULE 10

Forecasting: the financial model

Supports: Financial Modelling · Corporate Finance · Entrepreneurship

IN THE GAME

Under the live report of your own factory (Normal mode) the **Financial model** projects 12 months from the last 30 days, in three scenarios.



Financial model — a 12-month forecast from the last 30 days

BREAK-EVEN	MARGIN PER UNIT	FIXED COSTS	PAYS BACK
31 units / day	**$241.78**	**$7,378 / day**	**35 months**
sold at full price to cover fixed costs	price $527.64 – $285.85 to make	wages, upkeep, marketing	the $6.66M you built it with

SCENARIO	MONTH 1	MONTH 3	MONTH 6	MONTH 12	PROFIT IN 12 MONTHS
Bad	+$64.7K	+$68.5K	+$70.4K	+$70.4K	+$833K
Expected	**+$192K**	**+$197K**	**+$199K**	**+$199K**	+$2.38M
Good	+$265K	+$271K	+$274K	+$274K	+$3.27M

Expected keeps today's output and prices; as marketing grows your reach, more of what you make sells at the full price instead of to wholesalers. Bad: prices 8% lower, costs 5% higher, 15% fewer buyers. Good: prices 5% higher, 15% more buyers. Profit is before tax.

Break-even, margin per unit, fixed costs, payback and the 12-month forecast in three scenarios.

THE FINANCE

A model turns assumptions into a forecast. **Scenario analysis** asks "what if?" for a few coherent sets of assumptions (bad, expected, good); **sensitivity** asks how much the result moves for one input. The **payback period** is a simple, popular measure of risk: how long until the money invested comes back. It ignores what happens after payback and the time value of money (Module 11 fixes that).

- **Expected** keeps today's output and prices; as marketing grows reach, more units sell at full price instead of to wholesalers.
- **Bad**: prices 8% lower, costs 5% higher, 15% fewer buyers. **Good**: prices 5% higher, 15% more buyers.

FORMULA

Monthly profit = 30 × (full-price units × price + wholesale units × 0.88 × price − units × variable cost − fixed costs)
Payback (months) = amount invested ÷ monthly profit

WORKED EXAMPLE

The model above: the factory cost $6.66M of plant and earns about $192K a month in the expected case, so payback = 6.66 ÷ 0.192 ≈ **35 months**. In the bad case profit falls to about $65–70K a month, a 66% drop from prices 8% lower and costs 5% higher: operating leverage at work (Module 5). The forecast is **positive in every scenario**, so the project survives the bad case.

TRY IT IN THE GAME

Change your supplier or marketing and come back a week later: the model updates from the new 30 days.

MODULE 11

Capital budgeting and valuation

Supports: Corporate Finance · Valuation · Investment Appraisal

IN THE GAME

Every big decision costs money now for money later: a ship, a factory, an upgrade (+25% output, up to three levels), research. The company rating values your whole business by a **discounted cash flow** (DCF): five years of its cash flow discounted at the Meridia central-bank rate plus 6%, a terminal value, plus what you own today.

THE FINANCE

A dollar today is worth more than a dollar next year, because it could earn interest meanwhile. **Net present value** discounts each future cash flow back to today and subtracts the cost; a project with positive NPV creates value. The discount rate reflects the risk. DCF valuation applies the same idea to a whole company: the value of a business is the present value of the cash it will produce.

FORMULA

$PV = CF \div (1 + r)^t$ · $NPV = -\text{investment} + \Sigma\, CF_t \div (1 + r)^t$

Game DCF = net worth + $\Sigma_{t=1..5}\, CF_t \div (1 + r)^t + 6 \times CF_5 \div (1 + r)^5$, with r = Meridia rate + 6%

WORKED EXAMPLE

The first line upgrade on this Workshop costs about $1.47M (22% of its plant; later levels 26% and 30%) and adds 25% output: 18 units a day. At a $241.78 margin, if they sell at full price, that is $4,352 a day, $1.59M a year. At r = 9.5% for 5 years: $PV = 1.59 \times (1 - 1.095^{-5}) \div 0.095 = 1.59 \times 3.839 =$ **$6.10M**; NPV = 6.10 − 1.47 = **+$4.6M**. If the extra units only reach wholesalers (margin $178.47), cash flow falls to $1.17M a year and NPV to +$3.0M: still positive.

TRY IT IN THE GAME

Before an upgrade, check whether your markets can absorb 25% more output; if not, add a market first.

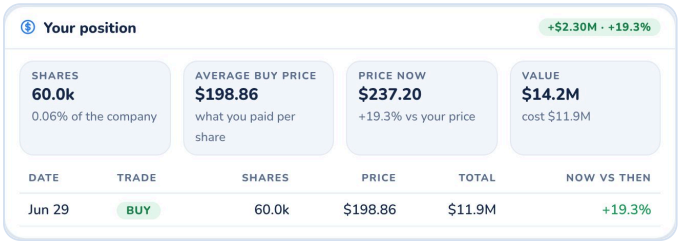
MODULE 12

Shares and the stock market

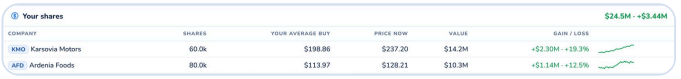
Supports: Investments · Corporate Finance · Equity Valuation

IN THE GAME

All 79 companies are listed. Their prices follow a **fair value** built from real earnings, growth, interest rates, reputation and the CEO, plus news-driven sentiment and daily noise. During the day prices move hour by hour. Dividends are paid each quarter from profit. In Pro mode, owning 5% gives a vote, 20% lets you bring proposals and 50% gives control.



Your position: shares, average buy price, price now, value and every trade.



Market → Stocks: all your shares with your average buy price and gain.

THE FINANCE

A share is a claim on future earnings. The **P/E ratio** is the price investors pay for one dollar of annual earnings; it rises with expected growth and falls when interest rates rise (future earnings are worth less today). Big orders move the price (**market impact**), so large positions are built and sold slowly. Your **average buy price** (cost basis) decides your gain or loss.

FORMULA

P/E = 17 + 22 × growth − 1.5 × (rate − 3.5) + reputation and CEO effects, between 6 and 30

Fair value per share = earnings per share × P/E (never below 45% of book value)

Your price impact ≈ 10% × √(shares ÷ shares outstanding) · Gain = shares × (price − average buy)

WORKED EXAMPLE

A company earns $12.90 a share with 0% growth in a country whose rate is 4.5%: P/E = 17 − 1.5 = 15.5, fair value ≈ $200. You buy 60,000 shares at an average of $198.86. Price now $237.20: gain = 60,000 × $38.34 = **$2.30M** (+19.3%).

TRY IT IN THE GAME

Buy a small stake, then raise interest rates as host in an online match (Rate hike) and watch P/Es fall.

MODULE 13

Technical analysis

Supports: Investments · Behavioural Finance · Trading

IN THE GAME

Every Stock tab has a candlestick chart (daily or weekly candles, the dashed last candle is today, moving live) with indicators you switch on and tools to draw on it. Your average entry price and every futures entry are highlighted bands, with the liquidation price as a dotted red line. B and S mark your own trades.



KMO with MA 20, MA 50, Bollinger bands, volume, RSI and MACD, a drawn trend line (+25.5%), a price line, your share entry and two futures entries.

THE FINANCE

A **candle** shows open, high, low and close. Indicators summarise price history: **moving averages** smooth it (price above a rising MA 20 = short-term uptrend; MA 20 crossing above MA 50 = "golden cross"); **Bollinger bands** sit two standard deviations around MA 20 (price at the top band is stretched); **RSI** above 70 is often overbought and below 30 oversold; **MACD** crossing above its signal line is a momentum buy signal. In this game prices are driven by fundamentals plus random noise, so indicators describe trends; they do not predict news. Treat them as tools for timing and risk, not as proof.

FORMULA

MA_n = average of the last n closes · $EMA_t = k \times \text{price} + (1 - k) \times EMA_{t-1}$, $k = 2 \div (n + 1)$

Bollinger = MA 20 ± 2σ · RSI = 100 − 100 ÷ (1 + average gain ÷ average loss), 14 days

MACD = EMA 12 − EMA 26 · signal = EMA 9 of MACD

WORKED EXAMPLE

KMO trades at $239.65, above both MA 20 and MA 50, and both averages are rising: an uptrend. RSI sits around 70: stretched, near overbought. A trader might hold but not add, and draw a price line at the last swing low as a stop. The trend line drawn on the chart measures the climb along the trend: +25.5%.

TRY IT IN THE GAME

Turn on RSI and MACD, use Measure across the last trend, and draw a price line where you would sell.

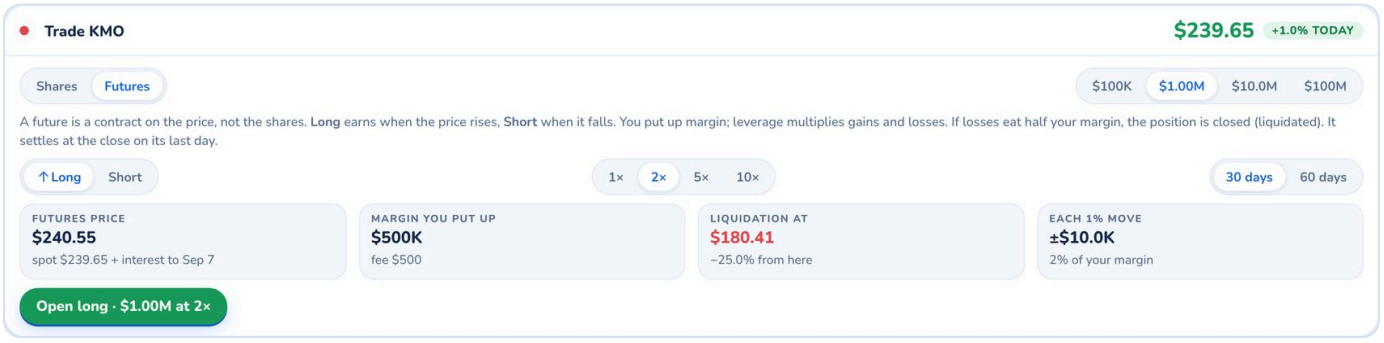
MODULE 14

Futures and leverage

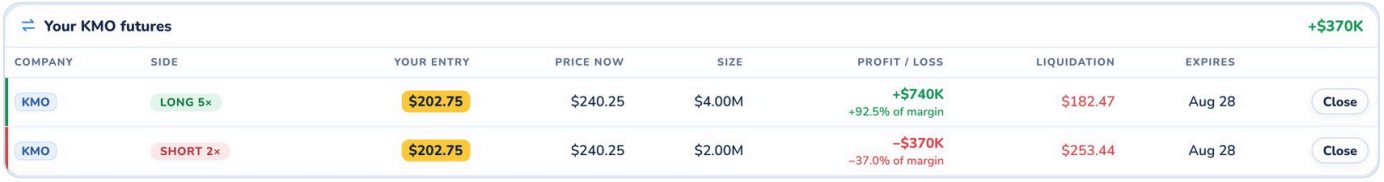
Supports: Derivatives · Risk Management · Investments

IN THE GAME

Market → **Live trading**, or the ticket on any Stock tab. Choose Long or Short, the size, leverage 1×, 2×, 5× or 10× and a 30- or 60-day contract. You put up margin; a 0.05% fee is charged on opening and closing. Positions are valued at the live price, settle at the close on their last day, and are closed automatically (liquidated) if losses eat half the margin. No player may hold more than 3% of a company's market value in its futures.



The futures ticket: futures price, margin, liquidation price and what each 1% move is worth.



Two KMO positions opened at the same price: the 5× long gained +92.5% of its margin, the 2× short lost 37%.

THE FINANCE

A **future** is a contract to trade at a fixed price on a later date. Its fair price is the spot price plus the **cost of carry** (interest until expiry). **Long** profits when the price rises, **short** when it falls, so futures can also **hedge**: a short future offsets a fall in shares you own. **Leverage** means you control a large position with a small margin; it multiplies returns in both directions and brings **liquidation risk**.

FORMULA

Futures price F = spot × (1 + r × days ÷ 360), r = the company's central-bank rate

Margin = size ÷ leverage · Contracts q = size ÷ F · P&L = side × q × (price now − entry)

Liquidation price = entry × (1 − side × 0.5 ÷ leverage) (side = +1 long, −1 short)

WORKED EXAMPLE

KMO spot $239.65, rate 4.5%, 30 days: F = 239.65 × (1 + 0.045 × 30/360) = **$240.55**. A $1M long at 2×: margin $500K, q = 4,157, liquidation = 240.55 × (1 − 0.25) = $180.41 (−25%). Each 1% move = ±$10K = 2% of margin.

The positions above, both entered at $202.75 with KMO now at $240.25 (+18.5%): **long 5×, $4M**: q = 19,729, P&L = 19,729 × 37.50 = **+$740K** on $800K margin (+92.5%), liquidation $182.47. **Short 2×, $2M**: q = 9,864, P&L = −$370K on $1M margin (−37%), liquidation $253.44, only 5.5% away.

TRY IT IN THE GAME

Open a small 2× long and a small 2× short on the same company and watch them move in opposite directions. Then hedge: short futures on a company whose shares you own.

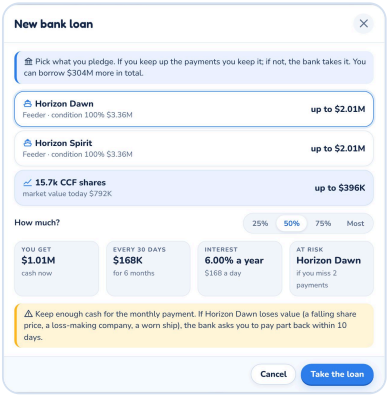
MODULE 15

Debt, collateral and default

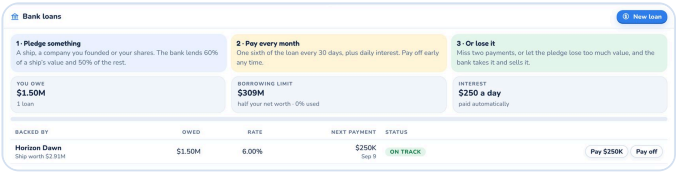
Supports: Banking · Credit Analysis · Corporate Finance

IN THE GAME

Finance → **Bank loans**. Every loan is secured: pledge a ship (the bank lends 60% of its resale value), a company you founded (50% of book value) or your shares in one company (50% of market value). Total borrowing may not pass half your net worth. One sixth of the loan is due every 30 days (a 6-month term), plus daily interest at the Meridia policy rate + 2.5 (ship), + 3 (company) or + 2 (shares) points.



A new loan against a Feeder: $1.01M now, $168K every 30 days, 6% a year.



Your loans: what backs each one, what you owe, the next payment and its status.

THE FINANCE

A **secured loan** gives the lender a claim on an asset. The **loan-to-value** (LTV) ratio keeps a safety cushion between the debt and the asset's value. If the asset falls in value, the lender issues a **margin call**; if the borrower **defaults**, the lender **repossesses** the collateral and sells it, keeping what it is owed. Default also damages your credit: in the game, no new loans for 60 days and 12 reputation points lost.

- **Missed payments**: if you lack cash on a due date the payment is missed; two missed payments and the bank takes the collateral.
- **Margin call**: if collateral × LTV × 1.3 falls below what you owe, you have 10 days to pay down the gap.
- **Auction**: ships sell at 85% of resale value, shares at 90% of market value, companies at 80% of book. Anything above the debt comes back to you; a shortfall is written off.

FORMULA

Maximum loan = collateral value × LTV · Payment every 30 days = original loan ÷ 6 · Interest per day = owed × rate ÷ 360
Margin call when collateral × LTV × 1.3 < owed

WORKED EXAMPLE

Horizon Dawn has a resale value of $3.36M, so the bank lends up to 60% = $2.01M. Taking 50% of that, $1.01M at 6.0%: $168K every 30 days and $168 of interest a day. Borrowing $6M against KMO shares worth $13M (max $6.5M) is close to the limit: if KMO falls 8%, the shares are worth $12M, × 0.5 × 1.3 = $7.8M, still above $6M. If it falls 30%: $9.1M × 0.65 = $5.9M < $6M → margin call.

TRY IT IN THE GAME

Borrow against a ship and keep enough cash for the monthly payment. Read the status column: "On track", "Missed 1 of 2" or "Margin call".

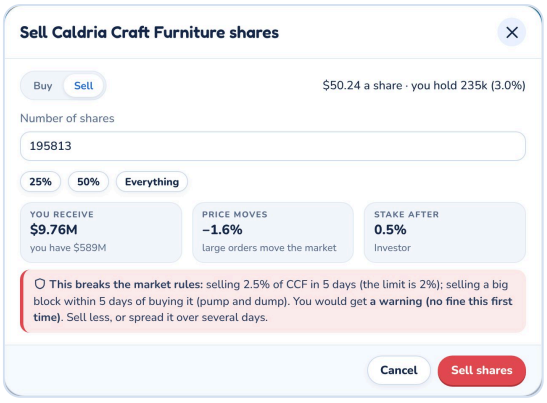
MODULE 16

Market integrity

Supports: Securities Regulation · Business Ethics · Market Microstructure

IN THE GAME

A market watchdog stops players from crashing prices for everyone. The rules are on the Finance screen with your record, and the share ticket warns you before a trade that breaks them.



The warning before a sale that breaks the rules.



The rules and your record: a warning, then a citation with a fine.

RULE	LIMIT
No share dumping	At most 2% of one company sold in 5 days; no single sale may push its price down more than 3%
No pump and dump	No selling a big block within 5 days of buying it
No cargo dumping	No selling more cargo in one port than 60% of its daily trade

THE FINANCE

Markets work only if prices are fair. **Dumping** crashes a price for everyone else; **pump-and-dump** drives a price up then sells into the rise; both are forms of **market manipulation**, which real securities law (for example the US Securities Exchange Act, or Indonesia's capital-market law enforced by OJK) prohibits. Regulators use escalating sanctions: a warning, fines that grow with repeat offences, and trading suspensions.

FORMULA

First offence in 90 days: warning · Then: fine = max($50,000, 15% of proceeds × (1 + 0.5 × earlier citations)), reputation −4
Third citation in 90 days: trading ban for 15 days (shares, futures and trade runs)

WORKED EXAMPLE

You hold 3% of Caldria Craft Furniture (7.26M shares). Selling 2.5% within 5 days of buying breaks two rules: the watchdog issues a warning. Selling another 0.3% the next day takes you to 2.8% in 5 days: citation, fine 15% of $2.13M proceeds = **$319K**. Spread over five weeks at 0.5% a week, the same sale is legal and moves the price less.

TRY IT IN THE GAME

Open Sell on a big holding and change the number of shares until the warning appears: that is the limit.

MODULE 17

Interest rates, inflation and the economic cycle

Supports: Macroeconomics · Monetary Economics · International Finance

IN THE GAME

Each of the ten countries has a central bank and a policy rate (from 3.0% in Norvaska to 6.5% in Demeron at the start). Prices and wages rise with inflation (2.5% a year by default). Rates raise companies' borrowing costs and lower share values. News, and in online matches the host, can shock the economy.

WORLD EVENT (HOST)	WHAT HAPPENS
Inflation shock	Prices and wages race up for about three months; central banks answer with higher rates
Stock market crash	Share prices fall 20–35% at once and stay weak for weeks
Oil price shock	Crude, gas and fuel jump about 50%; shipping gets dearer
Global recession / boom	Shoppers buy about 18% less / more for four months
Rate hike	Every central bank raises its rate 1.5 points today
Port strikes · Storm season	Three ports slow for two weeks · four storms slow every ship that crosses them

THE FINANCE

Central banks raise rates to cool inflation. Higher rates make borrowing dearer (your loans, Module 15), raise the futures cost of carry (Module 14) and lower the present value of future earnings, so P/E ratios fall (Modules 11–12). A **recession** cuts demand and spreads through supply chains; a **boom** does the opposite. Online hosts set the trend for a whole match: economy, share prices, commodity prices, inflation and volatility (Part 3).

FORMULA

Real return ≈ nominal return − inflation · Higher rate r → lower PV = CF ÷ $(1 + r)^t$ → lower P/E

WORKED EXAMPLE

A Rate hike moves Karsovia from 4.5% to 6.0%. KMO's P/E falls by 1.5 × 1.5 = 2.25 points, from 15.5 to 13.25: with the same $12.90 earnings, fair value falls from $200 to $171, about 15%. A 60-day future becomes dearer to carry: 6.0% × 60/360 = 1.0% above spot instead of 0.75%.

TRY IT IN THE GAME

In an online match as host, send a Rate hike and watch the stock indices and your loans' interest.

MODULE 18

Measuring performance: the company rating

Supports: Performance Management · Credit Rating · Strategic Management

IN THE GAME

Finance → **Company rating**: one score from 0 to 100 and a grade from AAA to D, recalculated every five days. It can decide online matches (Part 3).



A rating after 100 days: AAA 85.5. Strong profit, value and operations; growth still moderate.

PART	WEIGHT	WHAT IT MEASURES
Profit	25%	Money earned by ships, trade, factories and dividends in 60 days, against your start
Value (DCF)	20%	Net worth plus five years of discounted cash flow (Module 11)
Growth	10%	Net worth against your start
Ship operations	15%	Ship-days at work, on-time deliveries, ship condition
Factory operations	15%	Output against capacity, margin, crew mood, module health; no factory scores 30
Risk & conduct	15%	Reputation, borrowing against net worth, missed payments, market fines

THE FINANCE

A single financial number can mislead: net worth rewards a lucky trade as much as a well-run business. A **balanced scorecard** combines financial results with operations and conduct, so the best *operator* wins. Credit-rating agencies grade issuers the same way, from AAA (strongest) through BBB (the lowest investment grade) down to D (default).

FORMULA

Score = 0.25 profit + 0.20 value + 0.10 growth + 0.15 ships + 0.15 factories + 0.15 conduct

Grades: AAA 85+ · AA 75+ · A 65+ · BBB 55+ · BB 45+ · B 35+ · CCC 25+ · D below

WORKED EXAMPLE

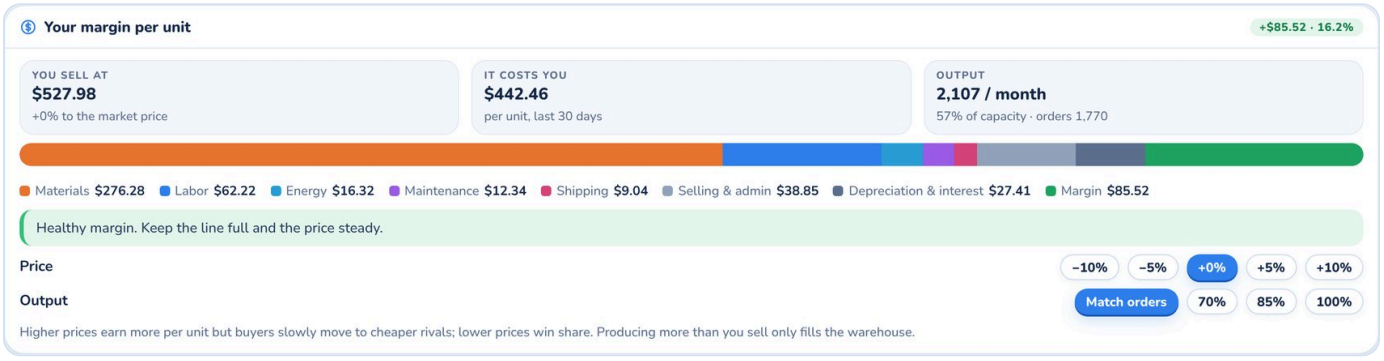
The rating above: 0.25 × 86 + 0.20 × 100 + 0.10 × 55 + 0.15 × 86 + 0.15 × 90 + 0.15 × 80 = 21.5 + 20 + 5.5 + 12.9 + 13.5 + 12 = **85.4** → AAA. The weakest part is growth (55): the panel advises reinvesting profit. A team that skipped factories would score 30 there and lose about 9 points.

TRY IT IN THE GAME

Read the two tips under your rating; they name your two weakest parts. Fix one and check again in five days.

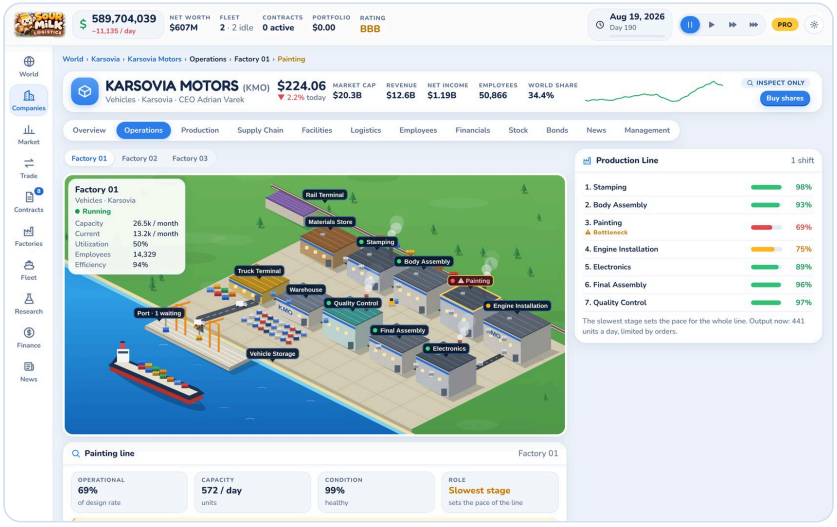
Pro mode: running a full company

In Pro mode your own company is planned like a real one: suppliers, production targets, staff, prices and markets are separate decisions with a benefit, cost, risk and time shown before you commit. The Overview of any company you run opens with **Your margin per unit**: what one unit sells for, what it costs line by line, and two dials.



Margin per unit in Pro mode: price, cost per unit, output, the cost split and the Price and Output dials.

- **Price** (−10% to +10% against the market): buyers choose suppliers by price, so a higher price earns more per unit but loses market share over the coming months; demand moves roughly with (average price ÷ your price)$^{2.5}$.
- **Output** (match orders, 70%, 85%, 100% of capacity): producing more than you sell only fills the warehouse; producing less leaves fixed costs spread over too few units.
- While orders are still building, a new company's crew works short hours and its overheads grow with its sales, so a well-run new factory usually breaks even in its second month.



Pro operations: the production line building by building, its bottleneck, the supply chain and the decisions.



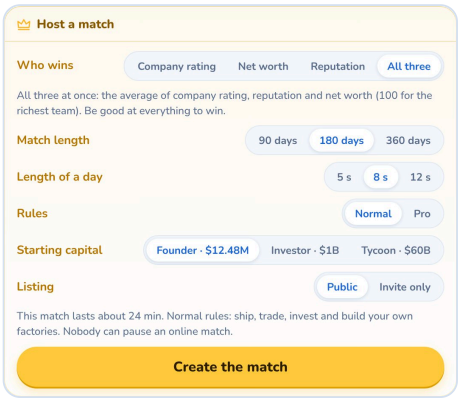
PART 3

Compete

Online matches: teams in one shared world, a market preview, host-chosen trends and four ways to win.

3.1 How an online match runs

Press **Play online** on the title screen. One player creates a match; others join with its five-letter code. Teams have 1 to 4 players; a friend joins your team with your team code. Spectators can watch any team live. Everyone shares one world: the same prices, contract board and companies, and the first team to take a contract gets it.



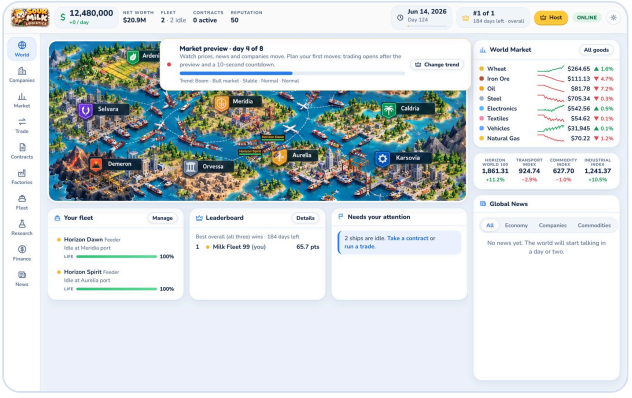
Creating a match: who wins, length, day length, rules, starting capital and listing.

WHO WINS	HOW IT IS SCORED
Company rating	The best rating (Module 18). Recommended.
Net worth	Cash, ships, shares, bonds, futures and companies, less loans.
Reputation	On-time deliveries; net worth breaks ties.
All three	The average of rating, reputation and net worth (net worth scores 100 for the richest team).

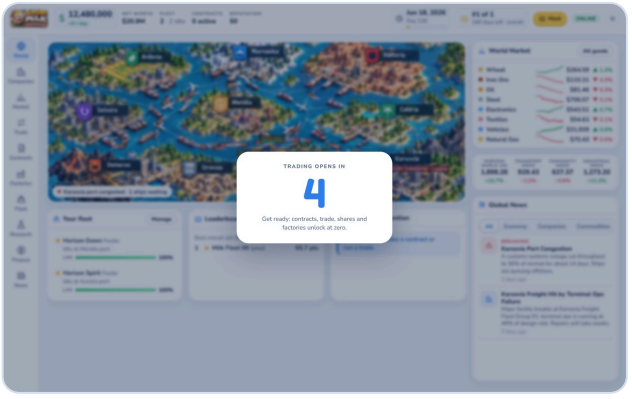
Matches last 90, 180 or 360 days, at 5, 8 or 12 seconds a day.

Market preview and countdown

When the host starts, the world first runs a **30-day market preview** (about 20 seconds). Everyone watches prices, news and companies move, but nobody can trade yet: a chance to study the market and plan. Then a **10-second countdown** opens trading for everyone at the same moment.

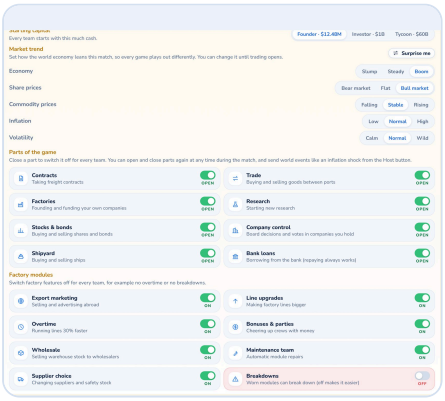


The market preview banner: day of preview, the trend and the host's Change trend button.



The countdown before trading opens.

3.2 Host controls



Host settings: starting capital, market trend, parts of the game and factory modules.

Market trend (until trading opens)

SETTING	OPTIONS
Economy	Slump · Steady · Boom
Share prices	Bear market · Flat · Bull market
Commodity prices	Falling · Stable · Rising
Inflation	Low 1% · Normal 2.5% · High 6%
Volatility	Calm · Normal · Wild

"Surprise me" picks a random trend, so every match plays out differently.

During the match

- **Parts of the game**: open or close Contracts, Trade, Factories, Research, Stocks & bonds, Company control, the Shipyard and Bank loans for everyone.
- **Factory modules**: switch off export marketing, line upgrades, overtime, bonuses & parties, wholesale, maintenance teams, supplier choice, or breakdowns.
- **World events**: one every 5 days at most (Module 17).

3.3 A 90-minute class session

TIME	ACTIVITY
0–15 min	Lecturer explains the goal ("All three" works well for a class) and the topic of the day. Teams of 3–4 form.
15–20 min	Host sets the trend (for example Slump + Bear market for a risk lesson). Start: teams study the 30-day preview.
20–65 min	Play a 90-day match at 8 s a day (12 min) or a 180-day match at 12 s (36 min). The host sends one or two world events tied to the topic.
65–80 min	Each team opens its Finance screen and explains its rating: strongest and weakest part, one decision that helped and one that hurt.
80–90 min	Debrief with the course map (4.1) and two practice questions (4.2).

Good pairings: **Rate hike** for Modules 11, 12, 14 and 17 · **Oil price shock** for Modules 1, 4 and 7 · **Stock market crash** for Modules 13–15 · factory modules off (no overtime, no maintenance) for Module 6.



PART 4

Practise

A course map, 24 practice questions with answers, and a glossary.

4.1 Course map

COURSE	MODULES
Introduction to Finance / Business	1, 2, 3, 5, 9, 12
Financial Accounting	2, 9
Managerial and Cost Accounting	1, 5, 6, 7, 10
Corporate Finance	1, 5, 10, 11, 15, 18
Financial Statement Analysis	9, 10, 18
Investments and Portfolio Management	3, 12, 13, 14
Derivatives and Risk Management	7, 14, 15
Banking and Credit	15, 17, 18
Financial Markets and Institutions	12, 14, 16, 17
Securities Law and Business Ethics	16
Macroeconomics and Monetary Economics	4, 17
Operations and Supply Chain Management	1, 4, 6, 7
Marketing and International Business	3, 8
Entrepreneurship	5, 8, 10, 11
Strategic and Performance Management	6, 18, Part 3

4.2 Practice questions

1. A Panamax sails 700 nm at 140 nm a day plus 2 port days. Fuel is $150 a barrel. What does the trip cost? *(Module 1)*
2. Why does the Ultra Large cost less per TEU per day than the Feeder, and when is the Feeder still the better choice? *(1)*
3. A $300,000 contract arrives 5 days late. What is paid, and what happens to reputation? *(2)*
4. A good costs $50 in a port with daily trade 4,000 units. What do you pay per unit for 800 units, without and with Trade Analytics? *(3)*
5. Copper cover falls from 20 to 10 days. What happens to its price and to an electronics maker's unit cost? *(4)*
6. Price $400, variable cost $250, fixed costs $6,000 a day. What is the break-even volume? *(5)*
7. Four modules run at 95%, 90%, 100% and 70%. At what pace does the line run, and which module do you fix? *(6)*
8. Name two costs of letting a crew stay unhappy. *(6)*
9. Budget saves $1,200 a day on materials, but buyers pay $900 a day less. Delays cost 1.5 × 3 days × 40 units × $200 a month. Is Budget worth it, and how does safety stock change the answer? *(7)*
10. A country buys 100 units a day at full reach. Your reach is 30%. How many units a day would High marketing add once reach reaches its target? *(8)*
11. Revenue $1.2M, materials $660K, other costs $360K. What are the gross margin and the operating margin? *(9)*
12. A factory cost $8M and earns $250K a month. What is its payback period, and what does payback ignore? *(10)*
13. An upgrade costs $3M and adds $1.2M a year for 5 years. At 10%, what is its NPV? *(11)*
14. Why does the DCF in the rating add a terminal value? *(11)*
15. Earnings per share $8, growth 10%, rate 3.5%. Estimate the game's P/E and fair value. *(12)*

16. You bought 10,000 shares at $50 and 5,000 at $62. What is your average buy price? *(12)*
17. Price is above MA 20 and MA 50, and RSI is 78. What does this suggest, and what does it not tell you? *(13)*
18. Spot $100, rate 6%, 60-day future. What is its price? *(14)*
19. A $2M short at 5× entered at $100. Where is liquidation, and what is the P&L if the price falls to $90? *(14)*
20. How can a short future protect shares you own? *(14)*
21. Shares worth $10M back a $4.5M loan. How far can they fall before a margin call? *(15)*
22. You have one citation in the last 90 days. You sell $3M of shares and break a rule. What is the fine? *(16)*
23. A central bank raises its rate from 4% to 5.5%. What happens to a P/E of 16 in the game? *(17)*
24. Team A: profit 90, value 80, growth 70, ships 40, factories 30, conduct 85. Team B: 70, 70, 60, 80, 85, 75. Who has the better rating? *(18)*

4.3 Answer key

1. Days = 700 ÷ 140 + 2 = 7; fixed 7 × $13,000 = $91,000; fuel 700 × 1.3 × $150 = $136,500; total **$227,500**.
2. Fixed cost per TEU falls with size ($5.40 against $13.75). The Feeder is better for small loads and small contracts, where the big ship would sail mostly empty.
3. $300,000 × (1 − 0.30) = **$210,000**; reputation −3.
4. Slip = 0.35 × 800 ÷ 4,000 = 0.07 → $50 × 1.08 = **$54.00**. With Trade Analytics slip = 0.049 → $50 × 1.059 = **$52.95**.
5. Scarcity pushes the copper price up; the electronics maker's material cost per unit rises and its margin shrinks unless it raises prices.
6. Margin $150; break-even = 6,000 ÷ 150 = **40 units a day**.
7. **70%**, the slowest module. Fix the 70% module first; improving the others adds nothing.
8. Lower pace (lost output) and quitting (empty jobs for five days, then a new worker). Also: worse rating (factory operations).
9. Net saving $1,200 − $900 = $300 a day, about $9,000 a month. Delays cost 1.5 × 3 × 40 × $200 = $36,000 a month, so Budget without stock loses about $27,000 a month. With safety stock that covers the delays, Budget saves about $9,000 a month minus the storage cost.
10. High targets 100% reach: 100 − 30 = **70 more units a day** (gradually, 3.5% of the gap a day).
11. Gross margin = (1.2 − 0.66) ÷ 1.2 = **45%**; operating margin = (1.2 − 0.66 − 0.36) ÷ 1.2 = **15%**.
12. 8 ÷ 0.25 = **32 months**. It ignores cash after payback and the time value of money.
13. PV = 1.2 × 3.791 = $4.55M; NPV = 4.55 − 3 = **+$1.55M**.
14. A business keeps earning after five years; the terminal value (6 × year-5 cash flow, discounted) captures that.
15. P/E ≈ 17 + 22 × 0.10 − 1.5 × 0 = 19.2; fair value ≈ 8 × 19.2 = **$153.60**.
16. (10,000 × 50 + 5,000 × 62) ÷ 15,000 = **$54.00**.
17. An uptrend that is stretched (overbought). It does not tell you about news or earnings, which drive this game's fair value.
18. 100 × (1 + 0.06 × 60/360) = **$101.00**.
19. Liquidation = 100 × (1 + 0.5 ÷ 5) = **$110**. q = 20,000; P&L = −1 × 20,000 × (90 − 100) = **+$200,000** on $400,000 margin.
20. If the share price falls, the short gains about what the shares lose, so the total moves less (a hedge).
21. Margin call when value × 0.5 × 1.3 < 4.5 → value < $6.92M, a fall of about **31%**.
22. max($50,000, 0.15 × $3M × 1.5) = **$675,000**.
23. P/E falls by 1.5 × 1.5 = 2.25 points, to **13.75**.
24. A: 22.5 + 16 + 7 + 6 + 4.5 + 12.75 = 68.75 (A). B: 17.5 + 14 + 6 + 12 + 12.75 + 11.25 = **73.5** (A). **Team B**, thanks to running ships and factories well.

4.4 Glossary

Arbitrage buying in one market to sell in another at a higher price.

Break-even the volume at which profit is zero.

Bottleneck the slowest step, which sets the pace of a process.

Candlestick a bar showing open, high, low and close.

Collateral an asset pledged to secure a loan.

Contribution margin price minus variable cost per unit.

Cost of carry the interest that makes a future dearer than spot.

DCF valuing cash flows by discounting them to today.

Dumping selling so much at once that the price collapses.

Fixed cost a cost that does not change with activity.

Future a contract to trade at a set price on a later date.

Hedge a position that offsets the risk of another.

Leverage controlling a large position with a small amount of capital.

Liquidation forced closing of a position whose margin has run out.

LTV loan-to-value: the loan as a share of the collateral.

Margin call a demand to pay down a loan whose collateral has lost value.

NPV present value of future cash flows minus the investment.

Operating leverage profit swinging more than sales because of fixed costs.

P/E ratio share price divided by earnings per share.

Payback period time for a project's cash to repay its cost.

Reach the share of a market's buyers who know your product.

Repossession the lender taking the collateral after default.

RSI relative strength index, 0–100, a momentum gauge.

Safety stock extra materials held against late deliveries.

Slippage the price moving against a large order.

Stock cover days of stock left at current demand.

Wholesale selling in bulk to traders at a discount.



Now go and play.

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